



Short communication

Occurrence and characterization of genetic determinants of β -lactam-resistance in *Escherichia coli* clinical isolatesRituparna Tewari^{a,b,*}, Feroze Ganaie^{a,c}, Nimita Venugopal^{a,b}, Susweta Mitra^{a,d}, Rajeswari Shome^a, Bibek R. Shome^{a,*}^a ICAR-National Institute of Veterinary Epidemiology and Disease Informatics, Bangalore, India^b Department of Microbiology, Jain University, Bangalore, India^c Department of Medicine, Division of Pulmonary/ Allergy/Critical Care, University of Alabama at Birmingham, Birmingham, AL, USA^d School of Basic and Applied Sciences, Dayananda Sagar University, Bangalore, India

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ABSTRACT

β -lactamase mediated resistance in *Escherichia coli* is a significant problem that requires immediate attention. Herein, we aim to characterize and understand the dynamics of the genetic determinants of β -lactam resistance (i.e. ESBL, AmpC, and MBL) in *E. coli*. Out of 203 *E. coli* isolates, genetic determinants of β -lactam resistance were identified in 50% (n = 101) of isolates. ESBL, AmpC, and MBL resistance determinants were detected in 78%, 40%, and 18% of isolates, respectively with *bla*_{CTX-M group 4} (48%), *bla*_{CMY} (40%), and *bla*_{SIM} (33%) as the most prevalent β -lactam resistance genes. Among these isolates, 45% harbored plasmid replicon types, with L/M (40%) and Y (33%) as the most dominant replicon types. Integrations were detected in 40% of such isolates, with Class-1 and Class-3 representing 62% and 55%, respectively. Overall, we observed high rate of genetic determinants of β -lactam-resistance in *E. coli* isolates recovered from patients in clinical settings. The co-occurrence of antimicrobial resistance genes and mobile genetic elements in a high percentage of isolates is a major concern and relates to complex resistance mechanisms. To combat the serious threat of antimicrobial resistance, it is imperative to develop strategies for robust surveillance and understand the molecular basis of resistance acquisition and transmission.

1. Introduction

For decades, the empirical treatment of *E. coli* infections has been more relayed on the β -lactam group of antibiotics, and the widespread use of these agents has resulted in the emergence of antibiotic-resistant strains (Thakuria and Lahon, 2013). The emergence of extended-spectrum β -lactamase (ESBL), AmpC β -lactamase (AmpC), and metallo β -lactamase (MBL) producers pose a serious clinical crisis and a public health concern. Of note, the increasing occurrence of multidrug-resistant (MDR) *E. coli* brings in an undesirable quandary in the treatment options (Maurya et al., 2016; Oberoi et al., 2013). The heightened use/misuse of antibiotics in human medicine, agriculture, and veterinary coupled with the transmissibility of resistance determinants mediated by plasmids, transposons, and gene cassettes in integrons, are some critical factors contributing to the rise of antibiotic-resistant bacterial pathogens (Cergole-Novella et al., 2011).

The phenotypic expression of ESBLs is controlled by specific

mutations in certain genes such as *bla*_{TEM}, *bla*_{SHV}, and *bla*_{CTX-M}. The mutations alter the amino acid configuration of the encoded proteins, thus conferring the ESBLs with the ability to hydrolyze a wider spectrum of β -lactam antibiotics, including penicillin and oxyimino-cephalosporins (Jena et al., 2017). AmpC β -lactamases, another group of clinically significant cephalosporinases, mediate drug resistance either through a specific mutation in the bacterial chromosome or by a transmissible element, mainly plasmid (Oberoi et al., 2013). Similarly, MBL-producing *E. coli* are being reported with increasing frequency from several parts of the world and have emerged as the most clinically effective carbapenem-resistance mechanism (Meletis, 2016). MBLs are encoded either by chromosomal genes or by heterologous genes acquired by horizontal gene transfer (HGT). Even though MBL-mediated resistance in *Enterobacteriaceae* represents a serious remedial challenge, the clinical data concerning their prevalence and genetic diversity are insufficient, especially from India (Frost et al., 2005).

Systematic monitoring of antibiotic resistance at the local, national,

* Corresponding authors at: ICAR-National Institute of Veterinary Epidemiology and Disease Informatics, Bangalore, India.

E-mail addresses: tewarirituparna@gmail.com (R. Tewari), brshome@gmail.com (B.R. Shome).

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Table 1Isolation of *E. coli* from various human clinical specimens and their correlation with the genetic determinants of β -lactam-resistance.

Specimen	No. of GNB isolates	No. of <i>E. coli</i> isolates	^a No. of GBLR- <i>E. coli</i>	^b ESBL_gd+	^c AmpC_gd+	^d MBL_gd+
KIMS H & RC						
Pus	27	21	11	11	5	2
Urine	31	23	15	9	9	3
Sputum	22	12	7	5	4	1
Fluid	9	4	4	3	3	0
Blood	11	3	5	5	2	0
Total	100	63	42	33	23	6
Gokula Metropolis						
Pus	20	16	6	5	1	1
Urine	25	23	11	8	1	2
Sputum	32	19	9	7	6	1
Fluid	15	10	6	6	1	0
Blood	8	6	2	2	0	1
Total	100	74	34	28	9	5
Kanva Diagnostics						
Pus	25	16	4	3	1	0
Urine	30	26	7	4	3	3
Sputum	18	8	7	5	2	3
Fluid	12	7	4	4	0	0
Blood	15	9	3	2	2	1
Total	100	66	25	18	8	7

E. coli strain ATCC 25922 (American Type Culture Collection, Manassas, VA, USA) was used as a positive control for identifying *E. coli*. ^a*E. coli* isolates positive for the genetic determinants of β -lactam-resistance, herein collectively referred to as “genetically β -lactam resistant *E. coli*” (GBLR-*E. coli*). ^bESBL_gd+: *E. coli* isolates positive for the ESBL genetic determinants. ^cAmpC_gd+: *E. coli* isolates positive for the AmpC genetic determinants. ^dMBL_gd+: *E. coli* isolates positive for the MBL genetic determinants. GNB, Gram-negative bacteria.

and international levels is an integral part of the control strategy. Infections with these ESBL-/AmpC-/ MBL-producing organisms continue to be associated with limited therapeutic options, resulting in higher mortality and morbidity rate and the high cost of treatment (Zilahi et al., 2016). Herein, we screened and characterized the chromosomal and plasmid-mediated genetic determinants of ESBL-, AmpC-, and MBL-resistance in *E. coli* clinical isolates collected from diagnostic and tertiary healthcare centers.

2. Materials and methods

2.1. Collection of isolates and laboratory processing

In this laboratory-based, cross-sectional study, a total of 300 consecutive, non-repetitive, clinical Gram-negative bacterial isolates recovered from various clinical specimens (i.e. pus, urine, sputum, blood, wound, ascitic, and pleural fluid) were investigated. The samples were collected from two diagnostic laboratories and a tertiary care referral hospital located in South Bangalore. The isolates were re-characterized for the identification of *E. coli* following colony morphology and standard biochemical tests (i.e. Urease production, Catalase test, Motility, Voges Proskauer, Indole production, Carbohydrate fermentation tests, Methyl red and Citrate utilization) (Isenberg, 2004).

2.2. *E. coli*-specific multiplex polymerase chain reaction

Genomic DNA extracted from the bacterial isolates was subjected to multiplex polymerase chain reaction (mPCR) for specific identification of *E. coli* by simultaneous amplification of multiple gene targets – *lacY* (289 bp), *cyd* (398 bp), *phoA* (468 bp), *lacZ* (517 bp), and *uidA* (603 bp) (Table S1) as previously described by Modak et al., 2012. The confirmed *E. coli* isolates and their corresponding DNA samples were stored at -80°C until further use.

2.3. PCR-based screening of antibiotic resistance genes

All the confirmed *E. coli* isolates were screened for the most common 17 antibiotic resistance genes (Table S1) using several PCR approaches. A) Six uniplex PCR assays targeting ESBL genes (*bla*_{TEM}, *bla*_{SHV}, *bla*_{CTX-M} group 1, 2, 3, and 4) (Tewari et al., 2019; Kojima et al., 2005; Pitout et al., 2004). B) mPCR-1 targeting AmpC genes (*bla*_{FOX}, *bla*_{MOX}, *bla*_{ACC}, *bla*_{DHA}, *bla*_{EBC}, and *bla*_{CMY}) (Tewari et al., 2019; Perez-Perez and Hanson, 2002). C) mPCR-2 targeting MBL genes (*bla*_{IMP}, *bla*_{VIM}, *bla*_{GIM}, *bla*_{NDM-1}, and *bla*_{SPM}) (Tewari et al., 2019; Mendes et al., 2007).

2.4. Plasmid replicon typing and detection of integrons

E. coli isolates positive for the genetic determinants of β -lactam-resistance, herein collectively referred to as “genetically β -lactam resistant *E. coli*” (GBLR-*E. coli*) were subjected to PCR-based replicon typing to determine the clinically important plasmid incompatibility (Inc.) groups (Tewari et al., 2019). PCR was performed with 18 pairs of primers (Table S1) recognizing B/O, FIC, A/C, P, T, K/B, W, FIIA, FIA, FIB, Y, HI1, I1, Frep, X, N, HI2, and L/M replicon types in three mPCR assays according to Johnson et al., 2007.

mPCR assay targeting integrase genes of the *intI1*, *intI2*, and *intI3* (Table S1) were used to detect the presence of Class-1, -2, and -3 integrons in all GBLR-*E. coli* isolates (Cao et al., 2014). Known positive and negative controls were included in each PCR run.

3. Results and discussion

ESBL, AmpC, and/or MBL-producing pathogens, especially *E. coli*, are becoming more prevalent in hospitalized patients and outpatients around the world (Oberoi et al., 2013). Given the ability of *E. coli* to develop incessant mechanisms of antibiotic resistance, characterization of genetic elements is essential to gain insights into antibiotic resistance gene repertoire. Furthermore, it helps in understanding the antibiotic resistance mechanisms and the potential for the development of high-risk multidrug-resistant strains. Herein, we studied the prevalence and characterized the genetic determinants of β -lactam resistance in *E. coli*

Table 2

Distribution of genetic determinants of ESBL, AmpC and MBL resistance in *E. coli* isolates recovered from patients in different healthcare centers.

Genetic determinant of resistance	Resistance Type	<i>E. coli</i> isolates recovered from healthcare centers			Total
		KIMS H & RC	Gokula Metropolis	Kanva Diagnostics	
<i>bla</i> _{TEM}	ESBL	5	4	4	13 (16%)
<i>bla</i> _{SHV}		1	0	0	1 (1.3%)
^a <i>bla</i> _{CTX-M group 1}		5	9	3	17 (22%)
^b <i>bla</i> _{CTX-M group 2}		0	0	0	0
^c <i>bla</i> _{CTX-M group 3}		0	0	0	0
^d <i>bla</i> _{CTX-M group 4}		12	8	4	24 (30%)
<i>bla</i> _{TEM} + <i>bla</i> _{SHV}		2	0	1	3 (4%)
<i>bla</i> _{TEM} + <i>bla</i> _{CTX-M group 1}		1	3	3	7 (9%)
<i>bla</i> _{TEM} + <i>bla</i> _{CTX-M group 4}		2	1	1	4 (5%)
<i>bla</i> _{SHV} + <i>bla</i> _{CTX-M group 4}		2	2	1	5 (6%)
<i>bla</i> _{TEM} + <i>bla</i> _{SHV} + <i>bla</i> _{CTX-M group 1}		2	0	0	2 (3%)
<i>bla</i> _{TEM} + <i>bla</i> _{SHV} + <i>bla</i> _{CTX-M group 4}		1	1	1	3 (4%)
<i>bla</i> _{TEM} + <i>bla</i> _{SHV} + <i>bla</i> _{CTX-M group 1} + <i>bla</i> _{CTX-M group 4}		33	28	18	79 (100%)
<i>bla</i> _{FOX}	AmpC	2	0	2	4 (10%)
<i>bla</i> _{MOX}		3	2	0	5 (13%)
<i>bla</i> _{EBC}		2	0	0	2 (5%)
<i>bla</i> _{ACC}		0	0	2	2 (5%)
<i>bla</i> _{CMY}		6	3	3	12 (30%)
<i>bla</i> _{DHA}		3	2	1	6 (15%)
<i>bla</i> _{CMY} + <i>bla</i> _{DHA}		3	1	0	4 (10%)
<i>bla</i> _{MOX} + <i>bla</i> _{FOX}		2	0	0	2 (5%)
<i>bla</i> _{EBC} + <i>bla</i> _{MOX} + <i>bla</i> _{DHA}		2	1	0	3 (7%)
<i>bla</i> _{IMP}	MBL	23	9	8	40 (100%)
<i>bla</i> _{VIM}		1	1	2	4 (22%)
<i>bla</i> _{GIM}		1	0	1	2 (11%)
<i>bla</i> _{SIM}		0	0	1	1 (6%)
<i>bla</i> _{SPM}		2	1	1	4 (22%)
<i>bla</i> _{NDM-1}		0	0	1	1 (6%)
<i>bla</i> _{VIM} + <i>bla</i> _{SIM}		1	1	0	2 (11%)
<i>bla</i> _{IMP} + <i>bla</i> _{SPM}		0	1	0	1 (6%)
<i>bla</i> _{IMP} + <i>bla</i> _{NDM-1}		1	0	1	2 (11%)
<i>bla</i> _{VIM} + <i>bla</i> _{SIM}		0	1	0	1 (6%)
<i>bla</i> _{VIM} + <i>bla</i> _{NDM-1}		0	1	0	1 (6%)
<i>bla</i> _{IMP} + <i>bla</i> _{SIM}		0	1	0	1 (6%)
<i>bla</i> _{IMP} + <i>bla</i> _{NDM-1}		0	1	0	1 (6%)
<i>bla</i> _{IMP} + <i>bla</i> _{SIM} + <i>bla</i> _{NDM-1}		0	1	0	1 (6%)
Total		6	5	7	18 (100%)

ESBL, extended-spectrum β -lactamase; MBL, metallo β -lactamase. ^a*bla*_{CTX-M group 1} includes CTX-M-1, -3, -10 to -12, -15 (UOE-1), -22, -23, -28, -29 and -30; ^b*bla*_{CTX-M group 2} includes CTX-M-2, -4 to -7 and -20 and Toho-1; ^c*bla*_{CTX-M group 3} includes CTX-M-8; and ^d*bla*_{CTX-M group 4} includes CTX-M-9, -13, -14, -16 to -19, -21 and -27 and Toho-2 (Pitout et al., 2004). ^eDetection of family-specific plasmid-mediated AmpC β -lactamase genes.

clinical isolates. Out of 300 clinical Gram-negative bacterial isolates, a total of 203 *E. coli* isolates were identified. The maximum isolation rate of *E. coli* was from urine (35%, $n = 72$) and pus samples (26%, $n = 53$) (Table 1). Interestingly, 27% ($n = 21$), 32% ($n = 13$), and 44% ($n = 8$) of uropathogenic *E. coli* isolates were positive for ESBL, AmpC, and MBL genetic determinants, respectively (Table 1). The escalation of β -lactam resistance in uropathogenic *E. coli* isolates has been well appreciated across the globe. In a study from Spain, 92% of ESBL-producing *E. coli* strains were isolated from urine specimens (Ahoyo et al., 2007). Similarly, in a recent study from India, 72% of uropathogens were ESBL-producing *E. coli* (Gajamer et al., 2020). This is expected since clinical isolates of *E. coli* are predominantly responsible for UTI. (Ahoyo et al., 2007; Gajamer et al., 2018; Gajamer et al., 2020).

Molecular characterization of 203 *E. coli* isolates identified 50% ($n = 101$) as GBLR-*E. coli*, with 78% ($n = 79$) of them positive for ESBL genetic determinants. The increased prevalence of ESBL producers has been reported from other major hospitals in India as well (Jones et al., 2002; Gajamer et al., 2020). Over-dependence on the third-generation cephalosporins to cure most bacterial infections is one of the prime reasons for the high resistance rate to this category of antibiotics. Among all the ESBL genes, *bla*_{CTX-M group 4} (48%, $n = 38$) was the most predominant genotype (Table 2). Co-occurrence of multiple genes were identified in 30% ($n = 24$) of isolates with *bla*_{TEM} + *bla*_{CTX-M group 1} (9%, $n = 7$) representing the common combination. Remarkably, 4% ($n = 3$) of GBLR-*E. coli* isolates harbored *bla*_{TEM} + *bla*_{SHV} + *bla*_{CTX-M group 1} + *bla*_{CTX-M group 4} gene combination (Table 2). Several studies from India, Europe, Latin America, and other countries have documented similar results (Al-Algamy et al., 2009; Gajamer et al., 2020; Devanga Ragupathi et al., 2020; Dhillon and Clark, 2012; Welsh et al., 2006).

In India, AmpC-producing Enterobacteriaceae strains has emerged as a challenge in hospitalized patients (Oberoi et al., 2013). In our study, 40% ($n = 40$) of *E. coli* isolates carried AmpC genetic determinants. Studies from different parts of the country reported AmpC-producing *E. coli* ranging between 7 and 48% (Suranjana and Manjusri, 2005). Selection criteria of isolates, geographical distribution, and the sample size could contribute to this large variation between the studies. Among the AmpC genes, *bla*_{CMY} (40%, $n = 16$) was the most common gene detected followed by *bla*_{DHA} (33%, $n = 13$) and *bla*_{MOX} (25%, $n = 10$). Multiple genes were identified in 23% ($n = 9$) of isolates with *bla*_{CMY} + *bla*_{DHA} (10%, $n = 4$) as the common combination (Table 2). In a study by Yamasaki et al., 2010, AmpC genes were detected in 22.7% of the total PCR positive isolates; however, they belonged to *bla*_{DHA} and the *bla*_{CIT} families. In another study from India, *bla*_{CIT} and *bla*_{EBC} were the most common genes detected in AmpC-positive *E. coli* strains (Basavaraju and Muttaraju, 2016).

MBL-producing Gram-negative organisms have now been reported in many geographic regions (Meletis, 2016.). Their ability to rapidly disseminate can lead to poor outcomes when infection ensues a concern, and therefore, early laboratory detection is of great clinical importance (Peleg et al., 2005). In the present study, the prevalence of GBLR-*E. coli* positive for MBL genetic determinants was 18% ($n = 18$), which is higher than reported by Gupta et al., 2006 (7.5%); however, it is much lower than Navaneeth et al., 2002. Among the MBL genes, the most dominant genes were *bla*_{SIM} and *bla*_{IMP} (each 33%, $n = 6$). Co-occurrence of genes were identified in 22% ($n = 4$) of isolates with *bla*_{IMP} + *bla*_{SPM} + *bla*_{NDM-1} (11%, $n = 2$) representing the common combination (Table 2). Manoharan et al., 2010 from Christian Medical College, Vellore, reported *bla*_{VIM} gene in 75% of MBL producers. *bla*_{IMP} and *bla*_{VIM} genes represent the most common resistance traits in carbapenemase-producing Enterobacteriaceae, and these MBL genes are known to be distributed globally (Buchunde et al., 2012; Chika et al., 2017; Manoharan et al., 2010; Mansouri et al., 2014).

Plasmid-mediated HGT represents a key mechanism by which resistance genes disseminate among different bacterial populations. Therefore, monitoring the spread of plasmids is useful to follow the transmission of antimicrobial resistance genes from different

Table 3Plasmid replicon types and integrons detected in ^aGBLR-*E. coli* isolates.

Resistance type	Plasmid replicon types									Integrons	
	B/O	FIC	A/C	FIA	Y	II	HI1	N	L/M	Class-1	Class-3
ESBL	–	6	2	6	6	–	–	2	4	7	4
AmpC	3	4	–	–	–	3	–	–	2	3	9
MBL	–	–	3	–	1	–	–	–	1	3	2
ESBL+AmpC	–	2	–	1	6	–	1	–	8	7	4
ESBL+MBL	–	–	–	–	2	–	3	–	–	3	2
AmpC+MBL	–	–	–	–	–	1	–	–	1	–	–
ESBL+MBL + AmpC	–	–	–	–	–	–	–	–	2	2	1

^a *E. coli* isolates positive for the genetic determinants of β -lactam-resistance, herein collectively referred to as “genetically β -lactam resistant *E. coli*” (GBLR-*E. coli*). Plasmid replicon types were identified in 45% ($n = 45$) of GBLR-*E. coli* isolates. Co-occurrence of Class-1 and Class-3 integrons was observed in 17% ($n = 7$) of isolates.

Table 4Co-occurrence of integrons and plasmid replicons in GBLR-*E. coli* isolates.

Isolate No.	Healthcare Center	Integrons		Plasmid replicon types								
		Class-1	Class-3	B/O	FIC	A/C	FIA	Y	II	HI1	N	L/M
1	KIMS	+	–	–	–	–	–	+	–	–	–	–
2	H & RC	+	–	–	–	–	–	+	–	+	–	–
3		+	–	–	–	–	+	+	–	+	–	–
4		+	–	+	–	–	–	–	–	–	–	–
5		+	–	–	–	–	–	+	–	–	–	+
6		–	+	–	–	–	–	–	–	–	–	–
7		–	+	–	+	–	–	+	+	–	–	+
8		–	+	–	–	–	–	+	–	–	–	+
9		–	+	–	–	–	–	+	–	–	–	+
10		–	+	–	–	–	–	–	–	–	–	+
11	Gokula Metropolis	+	+	–	+	–	–	–	+	–	–	+
12		+	+	–	–	–	+	+	–	+	–	+
13		+	–	–	–	–	–	+	–	–	–	+
14		+	+	–	–	+	–	–	–	–	–	+
15		+	+	–	–	–	–	+	–	+	–	–
16		–	+	–	+	–	–	–	–	–	–	–
17		–	+	–	+	–	–	–	–	–	–	–
18		–	+	–	–	–	+	–	–	+	–	–
19		–	+	–	–	–	–	–	–	+	–	–
20	Kanva Diagnostics	+	–	–	+	–	–	+	–	+	–	–
21		+	–	–	+	–	–	+	–	+	–	–
22		+	+	–	+	–	–	–	–	–	–	+
23		+	–	–	–	–	–	–	–	+	–	–
24		+	+	–	–	–	–	+	–	+	–	–
25		+	+	–	–	–	–	–	–	–	–	+
26		+	–	–	–	–	–	–	–	–	–	+
27		+	–	+	+	–	–	–	+	–	–	–
28		+	–	–	–	+	–	+	–	–	–	–

(+) positive; (–) negative.

environments. We identified nine plasmid replicon types among 45% ($n = 45$) of GBLR-*E. coli* isolates. The most common plasmid replicon type identified was L/M (40%, $n = 18$) (Table 3 and Table S2). Further, 18% ($n = 8$) of isolates harbored ≥ 3 plasmid replicon types (Table S2). Among the *E. coli* isolates positive for ESBL resistance genes, 42% ($n = 33$) harbored plasmid replicon types. Likewise, 75% ($n = 30$) and 50% ($n = 9$) of *E. coli* isolates positive for AmpC and MBL resistance genes carried plasmid replicon types (Table S2). We identified F replicons (FIA, FIC) predominantly in *E. coli* isolates positive for ESBL- and AmpC resistance genes (Table 3). These findings are in accordance with those of other studies (Johnson et al., 2007). IncF replicons are widely distributed among *E. coli* strains and seem to be well adapted to this species (Boyd et al., 1996). Curiously, the A/C replicon was detected in isolates carrying ESBL- and MBL-resistance genes, although this replicon has been rare in Enterobacteriaceae, particularly in *E. coli* isolated from humans (Johnson et al., 2007). Interestingly, L/M replicon type was identified in isolates positive for ESBL-, AmpC-, and MBL-resistance genes, suggesting its wider presence in *E. coli*.

Integrons were detected in 40% ($n = 40$) of GBLR-*E. coli* isolates with Class-1 and Class-3 representing 62% ($n = 25$) and 55% ($n = 22$) of such isolates, respectively (Table 3). Co-occurrence of Class-1 and Class-3

integrons was observed in 17% ($n = 7$) of isolates (Table S2). We didn't find Class-2 integrons in any of our isolates. The prevalence of Class-1 and Class-3 integrons found in our isolates was comparable to the observations of previous studies (Cocchi et al., 2007). Among *E. coli* isolates positive for the genetic determinants of ESBL, AmpC, and MBL resistance, 32% ($n = 25$), 60% ($n = 24$), and 50% ($n = 9$) of isolates harbored integrons, respectively (Table S2). Integrons are widespread in *E. coli* and may be responsible for high resistance rates, as observed in this study. Interestingly, 28% ($n = 28$) of isolates carried both genetic elements (i.e., plasmid replicons and integrons) (Table 4). Odetoyin et al., 2017 identified 81.7% of isolates harboring both plasmid replicons and integrons. The co-occurrence of these genetic elements significantly contributes to the dissemination of antibiotic resistance in Enterobacteriaceae and has been associated with specific genes conferring resistance to β -lactams, quinolones, and aminoglycosides.

4. Conclusion

The genetic determinants of β -lactam-resistance in *E. coli* clinical isolates recovered from the clinical settings are present at a high rate. The occurrence of multiple resistance determinants in these isolates

relates to complex antimicrobial resistance mechanisms. The high percentage of plasmid replicons and integrons in our study isolates represents an alarming public health concern, as they can act as vectors to spread antibiotic resistance. The study advocates the necessity for continuous monitoring of resistance trends at the genomic level and enhancing the infection control of these pathogens in healthcare settings.

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Credit author statement

BRS and RT: Conceived and designed the experiments;
RT: Sample collection;
RT, NV, and SM: Performed the experiments;
FG and RT: Data analysis and interpretation;
FG and RT: Writing- Original draft, review, and editing.
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Declaration of Competing Interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.meegid.2022.105257>.

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